AQA 3.2.2 and 3.2.3 Tests for ions (required practical 4)

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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18 marks · about 25 minutes

Key information

- Group 2 cations: NaOH(aq) gives white $Mg(OH)_2$; H_2SO_4(aq) gives white $BaSO_4$ with Ba^{2+}
- NH_4^+: warm with NaOH(aq), damp red litmus turns blue; $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$
- CO_3^{2-}: dilute acid gives effervescence; CO_2 turns limewater cloudy
- SO_4^{2-}: add HCl-acidified $BaCl_2$(aq) — white precipitate; the acid removes CO_3^{2-} first
- Halides with HNO_3-acidified $AgNO_3$(aq): Cl^- white, Br^- cream, I^- yellow; F^- no precipitate

Section A: Short-answer questions

01 Which silver halide(s) react(s) with concentrated ammonia solution to form a colourless solution?

[1 mark]

A silver chloride only
B silver chloride and silver bromide
C silver bromide and silver iodide
D silver iodide only

02 Some 1.0 mol dm^{-3} solutions were mixed using equal volumes of each solution. Which pair of solutions would give the greatest mass of solid?

[1 mark]

A $Ba(OH)_2$ and $MgCl_2$
B $Ba(OH)_2$ and $MgSO_4$
C $Ba(OH)_2$ and NaCl
D $Ba(OH)_2$ and Na_2SO_4

03 In the test for ammonium ions, the red litmus paper used must be damp. Explain why.

[1 mark]

04 What is the correct observation when barium metal is added to an excess of water?

[1 mark]

A Forms a colourless solution only
B Forms a colourless solution and effervesces
C Forms a white precipitate only
D Forms a white precipitate and effervesces

05 The sulfates of the elements in Group 2 from Mg to Ba have different solubilities. State the formula of the least soluble of these sulfates. Give a use for this sulfate.

[2 marks]

Section B: Longer questions

06 This question is about ion testing.

06.1 Describe how a student could distinguish between aqueous solutions of potassium nitrate, KNO_3, and potassium sulfate, K_2SO_4, using one simple test-tube reaction.

[3 marks]

06.2 Describe how a student could distinguish between aqueous solutions of magnesium chloride, $MgCl_2$, and aluminium chloride, $AlCl_3$, using one simple test-tube reaction.

[3 marks]

07 Solution Y contains two different negative ions. To a sample of solution Y in a test tube a student adds

- silver nitrate solution
- then an excess of dilute nitric acid
- finally an excess of concentrated ammonia solution.

The observations after each addition are recorded in the table.

Reagent added to solution Y	Observation
silver nitrate solution	cream precipitate containing compound D and compound E
excess dilute nitric acid	cream precipitate D and bubbles of gas F
excess concentrated ammonia solution	colourless solution containing complex ion G

Give the formulas of D, E and F. Give an ionic equation to show the formation of E. Give an equation to show the conversion of D into G.

[6 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**B**	*Source: 7404/1 Jun 2025 Q17*	1
02	**B**	*Source: 7405/3 Jun 2018 Q31*	1
03	Ammonia gas must dissolve (in water) to form OH^- ions / an alkaline solution	*Source: Practice question (written for this set)*	1
04	**B**	*Source: 7405/3 Jun 2019 Q15*	1
05	$BaSO_4$ X-rays (of internal organs) / barium meal	*Source: 7404/1 Jun 2023 Q02.4*	2
06.1	Reagent: $BaCl_2$ / $Ba(NO_3)_2$ / $Ba(OH)_2$ (or names) KNO_3: colourless solution / no visible change K_2SO_4: white precipitate	*Source: 7405/1 Jun 2017 Q08* If reagent incorrect or blank CE = 0/3; CE if H_2SO_4 Ignore 'nothing happens'	3
06.2	Reagent: NaOH / other Group 1 hydroxide $MgCl_2$: white precipitate $AlCl_3$: white precipitate which dissolves in excess	If reagent incorrect or blank CE = 0/3 Alternative: Group 1 carbonate — white ppt with $MgCl_2$; white ppt and effervescence with $AlCl_3$	3
07	D AgBr E Ag_2CO_3 F CO_2 $2Ag^+ + CO_3^{2-} \rightarrow Ag_2CO_3$ $AgBr + 2NH_3 \rightarrow Ag(NH_3)_2^+ + Br^-$ (2 marks: 1 for $Ag(NH_3)_2^+$, 1 for equation)	*Source: 7405/1 Jun 2019 Q05.5* Ignore state symbols If D = AgCl, allow 2 marks for $AgCl + 2NH_3 \rightarrow Ag(NH_3)_2^+ + Cl^-$	6